



General Details

Assessor	Samuel Prieto Ayllon
Assessment Date	02/07/2025
Assigned Reviewer	Samuel Prieto Ayllon
Next Review Date	02/07/2026

Signed Off By	Oraib AlKheetanRojas
Signed Off On	19/05/2026
Sign Off Comments	

Operation Assessed	Wireless Network Radio – Persistent Systems MPU5 Operation
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Associated with specific area

 Basement 2

 B2 029 (Robotics Lab)

 CTP

Description of work area and/or activity assessed

Researchers and students use the Persistent Systems MPU5 mobile ad-hoc networking (MANET) radio to establish high-bandwidth, low-latency wireless links between ground robots, sensors, and operator control stations. The radio is typically worn on a vest or mounted on a robot chassis inside the Kinesis Lab, but may also be deployed outdoors for field testing. Typical tasks include configuring frequencies, swapping battery packs, attaching external antennas, and streaming data/video during experiments.

Reason for Risk Assessment

To document and control any health, safety, and equipment-damage risks introduced by actively using the MPU5 radio in research.

Overall Current Risk

Low Risk



Average Number of Persons Affected

2	 Employees
2	 Students
2	 Visitors

Current Rating	Hazard Information	Task Status
Low Risk	Electricity The MPU5 operates from external Li-ion battery packs or a regulated DC supply (7-36 V). Damaged cables, incorrect polarity, or exposed contacts could cause short-circuits, mild electric shock, or sparks during installation or battery changes.	0 0 0
	Measures Currently in place to prevent risk of injury	
	Use manufacturer-approved batteries and power leads only. Inspect connectors and insulation before each use; do not operate with frayed or wet cables. Follow the “power-on last, power-off first” rule when integrating into robot wiring. Restrict repairs to trained CTP personnel; users must not open the chassis.	

Potential Rating	Additional Controls Required
Low Risk	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Low Risk	Non Ionising Radiation The MPU5 can transmit up to 6 W aggregate RF power across multiple antenna chains, well above typical Wi-Fi devices. Prolonged close-body exposure near active antennas could exceed ICNIRP occupational limits, and mis-configured antennas may concentrate energy in unintended directions.	0 0 0
	Measures Currently in place to prevent risk of injury	
	Maintain at least 20 cm separation between active antennas and body whenever practical; use remote-mounted antennas for fixed installations. Operate within licensed frequencies and configured power levels appropriate to the test environment. Post RF-warning signage during high-power bench tests; limit room occupancy to essential personnel. Verify antenna connections are snug and rated for the band to prevent high VSWR and unintended emissions.	

Potential Rating	Additional Controls Required
Low Risk	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
Low Risk 	Fire The radio's hot-swappable Li-ion battery packs may overheat, ignite, or release flammable electrolyte if physically damaged, short-circuited, or charged with an incompatible charger. The chassis can also become warm during continuous 6 W transmission.	0 0 0
	Measures Currently in place to prevent risk of injury	
	Store and charge batteries in a fire-retardant Li-ion charging bag inside the designated battery cabinet. Keep a CO2 or clean-agent extinguisher within 5 m of the test bench. Enable the radio's built-in temperature alarms; cease transmission and allow passive cooling if surface temperature becomes uncomfortable to touch. Train users to recognise signs of battery swelling or venting and to follow lab battery-incident procedures immediately.	

Potential Rating	Additional Controls Required
Low Risk 	No Remedial Actions Entered If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.